

## Week 4) Book exercise 9.1)

We want to show that the K-means algorithm must converge after a finite number of iterations.

To this end, we first note that there is only a finite number of ways in which you can partition  $N$  data points into  $K$  classes/clusters.

Furthermore, each partition is associated with an objective function value

$$J = \sum_{n=1}^N \sum_{k=1}^K r_{n,k} \|x_n - \mu_k\|^2$$

as defined in section 9.1 of the book.

We now note that each two-step iteration of the K-means algorithm (by construction) must (weakly\*) decrease  $J$ .

\*Note, however, that it is possible that two partitions have identical values of  $J$ , i.e., that  $J$  stays the same before and after an iteration of the algorithm.

To avoid a cycle/loop in our algorithm in this case, we should take care to break ties

consistently (e.g., by always assigning each observation to the lowest possible indexed cluster in case of multiple equal distance means).

It now follows that the K-means algorithm will converge after a finite number of iterations.

### Book exercise 9.3)

From (9.10) we have  $p(\underline{z}) = \prod_{k=1}^K \pi_k^{z_k}$

where  $z_k$  is the  $k$ 'th entry on 1-of- $K$  representation/encoding  $\underline{z}$  (see section 9.2).

- This is really just another way of saying that

$$p(z_k = 1) = \pi_k$$

where  $0 \leq \pi_k \leq 1$ ,  $k=1, \dots, K$  and  $\sum_{k=1}^K \pi_k = 1$ .

From (9.11), we are further given that

$$p(x|\underline{z}) = \prod_{k=1}^K U(x|\mu_k, \Sigma_k)^{z_k}$$

We now have that

$$p(x) \stackrel{\text{Law of total probability}}{=} \sum_{\underline{z}} p(x|\underline{z}) p(\underline{z}) = \sum_{\underline{z}} \left( \prod_{k=1}^K U(x|\mu_k, \Sigma_k)^{z_k} \right) \left( \prod_{k=1}^K \pi_k^{z_k} \right)$$

(9.10) and  
(9.11)

$$= \sum_{\underline{z}} \left( \prod_{k=1}^K (U(x/\mu_k, \underline{\Sigma}_k) \mathbb{I}_k)^{z_k} \right)$$

$$= \sum_{k=1}^K U(x/\mu_k, \underline{\Sigma}_k) \mathbb{I}_k$$

In each " $\underline{z}$ "  
 we have exactly  
one entry  $z_k=1$   
 while all other  
 entries  $z_j=0$   
 for  $j \neq k$ .  
 (see the definition  
 of  $\underline{z}$  in section 9.2).

which gives us (9.7).



### Additional exercise)

Assume that  $X_1, X_2, \dots, X_N$  are jointly Gaussian with  $\text{Cor}(X_i, X_j) = 0$  for  $i \neq j$ .

We want to show that  $X_1, \dots, X_N$  are independent.

Let  $\underline{X} = [X_1, X_2, \dots, X_N]^T$ . By assumption, we have that  $\underline{X} \sim \mathcal{WN}_N(\underline{\mu}, \underline{\Sigma})$  ← Multidimensional Gaussian distribution

where  $\underline{\mu} = [\mu_1, \mu_2, \dots, \mu_N]$  denotes a vector of the means of  $X_1, X_2, \dots, X_N$ , respectively, and

$$\underline{\Sigma} = \begin{bmatrix} \sigma_1^2 & 0 & \dots & 0 \\ 0 & \sigma_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \sigma_N^2 \end{bmatrix}$$

denotes a covariance matrix with the variance  $\sigma_1^2, \sigma_2^2, \dots, \sigma_N^2$  of  $X_1, X_2, \dots, X_N$ , respectively.

Note that  $\underline{\Sigma}$  is a diagonal matrix with determinant  $|\underline{\Sigma}| = \prod_{n=1}^N \sigma_n^2$ ,

the inverse  $\underline{\Sigma}^{-1}$  is a diagonal matrix with entries  $\frac{1}{\sigma_1^2}, \frac{1}{\sigma_2^2}, \dots, \frac{1}{\sigma_N^2}$ .

and that  $(\underline{x} - \underline{\mu})^T (\underline{x} - \underline{\mu}) = \sum_{n=1}^N (x_n - \mu_n)^2$ .

We then have a density  $f_{\underline{x}}$  of  $\underline{x}$  (i.e., a joint density of  $x_1, x_2, \dots, x_N$ ) given by

$$\begin{aligned} f_{\underline{x}}(\underline{x}) &= \frac{1}{(2\pi)^{N/2}} \frac{1}{|\Sigma|^{1/2}} \exp\left\{-\frac{1}{2} (\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu})\right\} \\ &= \left(\frac{1}{(\prod_{n=1}^N 2\pi)^{1/2}}\right) \left(\frac{1}{(\prod_{n=1}^N \sigma_n^2)^{1/2}}\right) \exp\left\{\sum_{n=1}^N -\frac{1}{2\sigma_n^2} (x_n - \mu_n)^2\right\} \\ &= \prod_{n=1}^N \frac{1}{(2\pi\sigma_n^2)^{1/2}} \exp\left\{-\frac{1}{2\sigma_n^2} (x_n - \mu_n)^2\right\} \\ &= \prod_{n=1}^N f_{x_n}(x_n) \end{aligned}$$

where  $f_{x_n}(x_n)$  denotes the Gaussian density function of  $x_n$  for  $n=1, \dots, N$ .

It then follows that  $x_1, x_2, \dots, x_N$  are independent.

## Hybrid exercise: Conjugacy and Gaussian Distributions

- 1) A prior  $p(\theta)$  and posterior  $p(\theta|X)$  distribution are called conjugate if they belong to the same family of distributions (i.e., have the same functional form). This is a nice algebraic property (and can make it more transparent how a likelihood is updated given data  $X$ ).
- See also section 2 in the book.

2) A data set is formed by independent and identically distributed data when each observation  $x_n$ ,  $n=1, \dots, N$  is the realization of a random variable  $X_n$  and  $X_1, \dots, X_N$  all follow the same/identical distributions while being mutually independent.

3) Consider the expression of  $p(x)$  as given in (1) in the exercise. Note that the fraction on the right-hand side of the expression is fixed for all data points  $x_1, x_2$ , and  $x_3$ . Furthermore,  $\exp(-z)$  is a monotonically decreasing, concave function in  $z$ . Thus,  $p(\underline{z})$  will decrease in

$$(\underline{z} - \mu)^T \Sigma^{-1} (\underline{z} - \mu) = z_1^2 + \frac{1}{2} z_2^2 \quad (*)$$

where  $z_1$  and  $z_2$  denote the first and second entry/dimension in  $\underline{z} = [z_1, z_2]^T$ .



It is now obvious that for points  $x_1$  and  $x_2$ ,  
 $p(x_1) < p(x_2)$ .

- This also makes sense intuitively, as  $x_1$  and  $x_2$  have the same euclidean distance to the mean  $\mu$ , but the "deviation" for  $x_2$  is "placed" in the second dimension with more variance (and thus more "likelihood").

Based on (1\*), we can further see that "proportionate" increases/decreases in the first and second dimensions "favor"  $p(\underline{z})$  by increasing the first dimension and decreasing the second dimension. To be specific, we then have that

$$p(x_1) < p(x_3), \text{ and}$$

$$p(x_3) > p(x_2).$$

- Again, this makes intuitive sense given the "variance" trade-off.
- Note, also, that the first and second dimensions of  $\underline{x}$  (or  $\underline{z}$ ) must be independent based on the "additional exercise" from this exercise class.



## Hybrid exercise: Nearest Neighbor Classifier)

To solve this exercise, we simply calculate the euclidean distance between  $x_6$ ,  $x_7$ , and  $x_8$  and all training points.

We then classify  $x_6$ ,  $x_7$ , and  $x_8$ , respectively, as the same class as their "nearest neighbour".

-I'm too lazy to do it. But it should really not be a problem for you.